# Project Management

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**Notes**

- Our subject code is CT 658.
- This document encorporates for both, electronics and computer department..
- If a question was asked in Regular Exam, it is kept in **boldened** form. If it was asked in back exam, it is in regular font.
- Months are marked as:
  - Ba: Baisakh
  - Jth: Jestha
  - Asa: Ashar
  - Shr: Shrawan
  - Bh: Bhadra
  - Ash: Ashwin
  - Ka: Kartik
  - Mng: Mangsir
  - Po: Poush
  - Ma: Magh
  - Ch: Chaitra

# Contents

# 1 Introduction

(2 Hours / 4 Marks)

## 1.1 Defintion of project and project management

1. Define project. [1] (`79 Ash`) [2] (**`79 Ch`**)
2. How is project different from the operational work? [2] (`82 Ka`)
   |→Differentiate between project and operation. [2+2] (79 Ch)
3. Define project management. [1.5] (**`81 Ch`**)
4. Why it project management important. [2] (**`81 Ch`**)
5. (Assumed) How can you relate the characteristics of project with your minor project? [4] (78 Ch)
6. Short notes on: Characteristics of project. [3] (`80 Ash`)

## 1.2 Project Objectives

1. What are the triple constraints of project? [3] (`82 Ka`)
2. Short notes on: Project tripple constaints [3] (**`81 Ch`**) [4] (**`79 Ch`**)

## 1.3 Classification of Projects

1. List out different bases of project classification. [5] (`79 Ash`)

## 1.4 Project Life Cycle

1. How do you identify the completion of project phases? [1.5] (**`81 Ch`**)
2. What are the phases in project life cycle? [2] (`80 Ash`)

# 2 Project Management Body of Knowledge

(4 Hours / 7 Marks)

1. What is PMBOK. [2] (**79 Ch**, 82 Ka)

## 2.1 Understanding of project environment

1. What do you mean by Internal, External and Task/operation environment in project? [2+2+2] (79 Ash)

2. How the external environmental influences on ICT project? [3] (**79 Ch**)

## 2.2 General management skill, effective and ineffective project managers

1. What do you mean by general management skill? [2] (**81 Ch**)

2. What are the essential interpersonal and managerial skills? [3] (**81 Ch**) [2+3] (80 Ash)

3. One of the famous actors, Bill Cosby, once said "I don't know the key to success but the key to failure is trying to please everybody." How relevant is this statement in the field of IT System Project Management? What are other traits of successful Project Manager? Explain. [6] (**80 Ch**)

4. Explain role and responsibilities of key project members. [6] (**78 Ch**)

5. Differentiate effective and ineffective project managers, at least with 5 different characteristics. [6] (81 Ash)

## 2.3 Management Knowledge

### 2.3.1 Essential interpersonal and managerial skills

1. What are the essential interpersonal & managerial skills to be successful project manager? [2+3] (82 Ka)

### 2.3.2 Energized and initiator

### 2.3.3 Communication, influencing, leadership, motivator

1. How the external environmental influence on ICT project? [3] (**79 Ch, 78 Ch**, 82 Ka)

### 2.3.4 Negotiation, problem solver, perspective nature, result oriented, global illiteracies

1. (Assumed) Short notes on: Problem in ICT Project. [3] (80 Ash)

### 2.3.5 problem solving using problem trees.

# 3 Portfolio and Project Management Institutes' Framework

(2 Hours / 4 Marks)

1. What is PMI Framework. [4] (79 Ash)

## 3.1 Portfolio

1. Define project portfolio management. [2] (**79 Ch**, 81 Ash)
2. Short notes on: Portfolio Management. [3] (**81 Ch**)

## 3.2 Project Management Office

1. Define project management office. [2] (80 Ash)

## 3.3 Drivers of Project Success

1. Discuss briefly the drivers for making project success. [4] (81 Ash)
2. What are the reasons for success of ICT projects? [3] (80 Ash)

## 3.4 Inhibitors of project success

# 4 Project Management

(4 Hours / 7 Marks)

## 4.1 Advantages of project management

## 4.2 Project management context as per PMI

1. Compare Project Manager versus Program Manager as per PMI. [4] (**80 Ch**)
2. Describe briefly about nine knowledge area under PMI. [6] (80 Ash)

## 4.3 Management

### 4.3.1 Characteristics of project life cycles

### 4.3.2 representative project life cycles

### 4.3.3 IT Product Development Life Cycle

### 4.3.4 Product Life Cycle and Project Life Cycle

1. Differentiate between product life cycle & Project life Cycle? [2] (**81 Ch**, 82 Ka) [3] (80 Ash)
2. What are the typical stages of Project Life Cycle, as per PMI? Write a brief note. [4] (81 Ash)

# 5 Project and Organization structure

(2 Hours / 4 Marks)

## 5.1 System view of project management

1. What is system view of project management? [2] (**81 Ch, 80 Ch, 79 Ch, 78 Ch**)
2. Explain about three sphere model of system management. [3] (**79 Ch**)[4] (**81 Ch, 80 Ch**)

## 5.2 Functional Organization, matrix organization

1. Describe types of organizational structure. [4] (**81 Ch**, 82 Ka) [6] (80 Ash)
2. Differentiate and compare among various organizational structures, with examples. [6] (**80 Ch**)
3. Describe about functional organization structure. [3] (**79 Ch**)
4. What are the disadvantages of projectized organizational structure? [2] (**79 Ch**)
5. Explain about Matrix organization and its type. [6] (79 Ash)

## 5.3 Organizational structure influences on projects

1. Explain, how organizational structures influence the IT projects? [3] (78 Ch)

# 6 Project Management Process Groups

(2 Hours / 4 Marks)

## 6.1 Project Management process

1. (Assumed) Write short notes on Role of Feasibility Study [3] (**81 Ch**)
2. Describe about project process groups. (**79 Ch**)
3. Explain the concept of project management processes and discuss how process groups overlap within a project. [5] (81 Ash)

## 6.2 Overlaps of process groups in a phase, mapping of project management process groups to area of knowledge

1. Explain overlaps of project process groups with clear diagram. [5] (78 Ch)

# 7 Project Integration Management

(4 Hours / 7 Marks)

1. What is Project Integration Management? [2] (**79 Ch**) [3] (**78 Ch**)

## 7.1 Develop project charters

1. What is project charter? [1] (**81 Ch**, 80 Ash)
2. Write short notes on: Project Charter. [3] (82 Ka)
3. What type of information should be included while making the project charter? [3] (79 Ash)
   |→Explain with example. [3] (**79 Ch**) [4] (**81 Ch**)
4. What are the essential information required to create Project Charter? [3] (**79 Ch**)

## 7.2 Develop preliminary project scope statement

## 7.3 Develop project management plan

## 7.4 Direct and manage project execution

## 7.5 Monitor and control project work

## 7.6 Integrated change control, close project, project scope management

1. Short notes on: Statement of Work (SOW) [3] (**81 Ch**)
2. Imagine you have got the role of Project Manager for a software project that will make personalized and secure access of your semester-end mark-sheet possible. This will relieve Exam Control Office for printing semester end marksheet rather students will browse and download semester end mark-sheet as pdf only through either using browser or ECD app. (80 Ch)
   - Prepare a brief project scope document for the above system project work. [7]
   - While making the process groups, what are the important considerations that you have to manage and take care of? [5]
3. Describe about project scope management process. [4] (80 Ash)
4. Define project scope with example. [2] (79 Ash)

## 7.7 Create Work Breakdown Structure

1. What is work breakdown structure? [1] (82 Ka)
2. Short notes on: WBS [3] (**81 Ch**)
3. How does WBS affect the work estimatae of the project duration/activity? Explain with example. [4] (82 Ka)

## 7.8 Scope Verification

## 7.9 Score Control

# 8 Project Time Management

(4 Hours / 7 Marks)

## 8.1 Activity definition, decomposition of activities, activity attributes

## 8.2 Activity sequencing, precedence relationship

## 8.3 Network diagram, precedence diagram method, arrow diagramming method

1. Consider the following project and answer the followings. [2+2+2+1+2] (82 Ka)

| Activity | Predecessor | Duration (week) |
|---|---|---|
| A | - | 2 |
| B | - | 6 |
| C | - | 4 |
| D | A | 3 |
| E | C | 5 |
| F | A | 4 |
| G | B, D, E | 2 |
| Total = | | 26 weeks |

- Draw the network diagram.
- Draw the forward and backward pass.
- Calculate the critical path with their respective activity.
- Calculate the total duration of the project.
- Find the total float of activity D, F.

2. Consider the following project and answer the followings. [2+2+1+2+2] (**81 Ch**)

| Activity | Predecessor | Duration (days) |
|---|---|---|
| A | - | 11 |
| B | - | 15 |
| C | A | 8 |
| D | A, B | 10 |
| E | B | 5 |
| F | C, D | 4 |
| G | E, F | 7 |
| Total = | | 60 days |

- Draw the network diagram.
- Draw the forward and backward pass.
- Calculate the total duration of the project.
- Find the total float on each activity.

3. Consider the following project and answer the followings. [10] (**80 Ch**)

| Activity | Predecessor | Duration (weeks) |
|---|---|---|
| A | - | 2 |
| B | A | 3 |
| C | A | 5 |
| D | B | 1 |
| E | A, B | 2 |
| F | C, D | 4 |
| G | E, F | 3 |

- Draw the activity network diagram.
- Draw the forward and backward pass.
- Calculate the total float and free float.
- Identify the critical activity and critical path.

4. Consider the following project and answer the followings.

| Activity | Predecessor | Duration (weeks) |
|---|---|---|
| A | - | 3 |
| B | A | 4 |
| C | A | 2 |
| D | B | 5 |
| E | C | 1 |
| F | C | 2 |
| G | D, E | 4 |
| H | F, G | 3 |

10 (81 Ash)

- Draw the network diagram.
- Find the critical path.
- Calculate the late start and late finish.
- Calculate the total float time of activity E, F.

4+2+3+3 (79 Ash)

- Prepare network diagram
- identify critical activity.
- find project duration.
- find EST, EFT, LST, LFT, TF, FF, IF and independent float.

5. Consider the following activity table: [3+3+2] (80 Ash)

| Activity | Predecessor | Duration (days) |
|---|---|---|
| A | - | 3 |
| B | A | 5 |
| C | A | 7 |
| D | B | 10 |
| E | C | 5 |
| F | D, E | 4 |

- Draw the network diagram.
- Find the critical path and activity.
- Calculate the project duration.

## 8.4 Activity resources estimating

## 8.5 Determining resource requirements

## 8.6 Schedule development and control

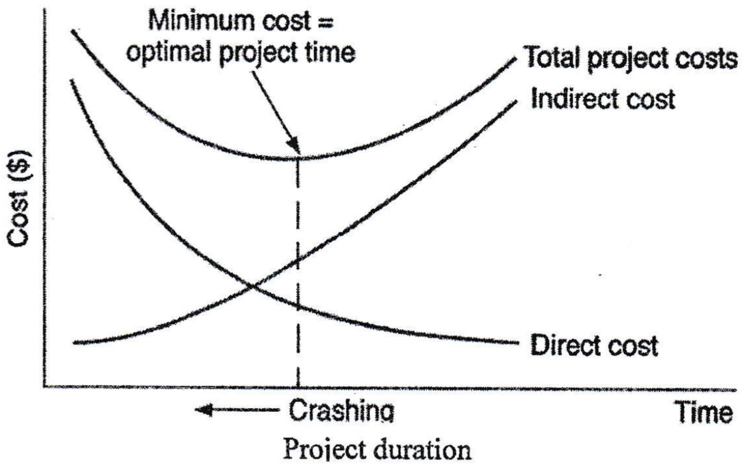
## 8.7 Principles of scheduling

### 8.7.1 milestones, forward pass, backward pass

### 8.7.2 Critical path method, critical chain technique

### 8.7.3 Gantt chart, schedule control

1. Project Crashing is termed as optimal point of trade-off between project cost and project time for any project while assessing delayed or over-run project. Do you agree or not? In any case justify with the help of right-side graph. [6] (**80 Ch**)



# 9 Project Cost Management

(4 Hours / 7 Marks)

## 9.1 Cost and project, cost management

## 9.2 Cost estimating, types of cost estimates, estimating process and accuracy

1. Write short notes on: COCOMO for ICT project. [3] (82 Ka)

## 9.3 Cost estimating tools

## 9.4 Cost budgeting, cost aggregation, deriving budget from activity cost

## 9.5 Cost control process, cost control methods, earned value management

1. Compare Project Cost management versus Earned Value management. [3] (**80 Ch**)

## 9.6 EVM benefits, variance analysis

1. What is EVM? [1] (**79 Ch**, 80 Ash) [2] (**81 Ch, 78 Ch**, 82 Ka)

## 9.7 Numerical

1. Suppose you have project with total budget Rs. 500,000/- scheduled to last for 8 months, you check your records and find that you have spent Rs. 300,000/- so far. The team has completed only 40% of the project work but when you check the schedule, it says they should have to complete 50% of the work. Now calculate EV, PV, SPL, CPI, EAC, and find the new estimate duration for above project. [1+1+1+1+1+1] (82 Ka)

2. Your project is scheduled for 2 years. Nine months into the project, while the total project budget is Rs 42,00,000. You have already spent Rs 16,50,000. CPI is 0.875. Calculate EV, EAC, ETC, VAC and is the project is behind the schedule or ahead schedule? [1+1+1+1+2] (**81 Ch**)

3. Suppose you are the project manager on an IOT project. The project is set to be completed in 10 months with an estimated cost of Rs 500,000. The project has been running for 5 months now, the team has spent Rs 220,000 and completed an amount of work worth Rs 255,000. Calculate CPI and SPI and Share your conclusion. [1+2+2+1] (79 Ch)

4. Calculate the schedule and cost variance for a project whose estimated budget is $100,000 and estimated time is 50 man months. 5 people were involved in the project. After 7 months, 60% of work was completed and $80,000 was expended. Also estimate the new cost and time for the project. [5] (**78 Ch**)

5. The project was scheduled to be completed in 12 months and budget as Rs 1,20,000. After 4 months, it was found that Rs. 50,000 has been spent and 40% of the work has been completed. [2+2+2] (82 Ash)

    - Calculate the cost variance and schedule vriance for the project.
    - Calculate the cost performance index and schedule performance index.
    - Calculate the estimated cost and time for completing the project if the project continues without any changes.

6. Your project is scheduled for 2 years. There are six different teams working on five major functional areas. Some teams are ahead of schedule while others are falling behind. There are more cost overruns in some areas but you have also saved costs in others. Due to all this, it is difficult to understand whether you are over or under budget. Nine months into the project, while the total project budget is Rs 42,00,000/- you have already spent 16,50,000/- CPI is 0.875/-. Can you perform EV analysis and budget forecast? and find the value of EV and VAC. Is the project over or under bduget? [2+1+2+1] (80 Ash)

7. Suppose you have a project that is scheduled to be completed in 10 days aat a budgeted cost of 100000/- at the end of day 6, you do analysis and you determine the job is 70% complete, and you have spent 65,000/-. (79 Ash)

   - What is the project's EV at Day 6? [1]
   - What is the project's budget at completion? [1]
   - Is the project ahead, behind schedule or on time? [2]
   - Is the project expected to complete on budget, under or over budget? [2]
   - What is the project's SPI nd CPI at Day 6. [2]

# 10 Project Quality Management

(3 Hours / 5 Marks)

## 10.1 Quality theories

1. How to improve quality in ICT project. [2] (82 Ka)

2. Define project quality. [2] (80 Ash)

## 10.2 Quality planning, project quality requirements, cost of quality, quality management plan

## 10.3 Quality assurance, quality audit, approach to quality audit

1. Quality is one the most important factors to be considered for effective delivery of project objectives. How quality assurance and quality control is implemented in order to deliver a successful project? [2] (**81 Ch**) [3] (**79 Ch**) [5] (**79 Ch**)

2. How is quality assured in software industry? [2] (79 Ash)

## 10.4 Quality control process, control chart, pareto charts, testing of IT system, the test life cycle

1. How the control charts used to monitor the quality of project result? [3] (82 Ka)

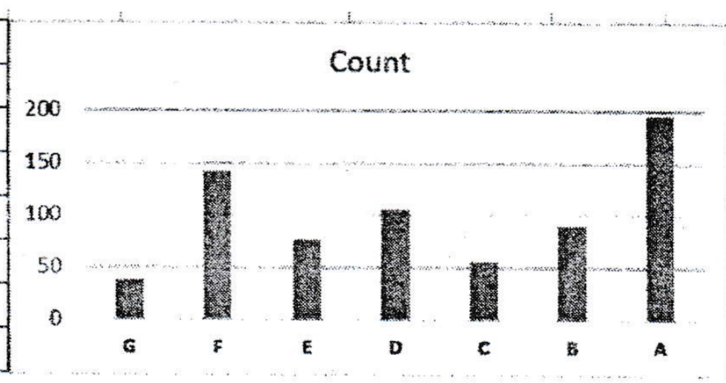
2. Write short notes on: TQM. [3] (**81 Ch**)

3. Explain 80/20 rule of the pareto principle with exmaple. [3] (80 Ash)

4. Quality is one the most important factors to be considered for effective delivery of project objectives. How quality assurance and quality control is implemented in order to deliver a successful project? [2+3] (**81 Ch**)

5. Consider below data about customer complaint on e-commerce portal and their srevices. Draw a free-hand sketch of Pareto diagram after having pareto analysis with clearly showing all necessary calculations. As per Pareto principles, which are the first three complaints should be addressed first? What will be the coverage of issues by these solving such three issues? [6+2] (**80 Ch**)

| Code | Complaint Types | Count |
|---|---|---|
| G | Unexpected fees | 37 |
| F | Unresponsive customer support | 142 |
| E | Lack of security | 76 |
| D | Poor display on mobile | 105 |
| C | Wrong product delivery | 55 |
| B | Product different than display | 90 |
| A | Slow website speed | 195 |




6. ISO/IEC 5055:2021 is an ISO standard for measuring the internal structure of a software product on four business-critical factors: Security, Reliability, Performance Efficiency and Maintainability. If you are building a typical software and associated support system of accounting and financial audit

record management system for academic Institutes and universities, how you will be using ISO/IEC 5055:2021 for ensuring quality of your system? [8] (81 Ash)

7. Differentiate quality assurance and quality control. [2] (79 Ash)

# 11 Project Communication Management

(3 Hours / 5 Marks)

## 11.1 Importance of communication management

1. What is project communication management? [1] (**81 Ch**)
2. Why communication management is important in IT project? [2] (**79 Ch, 78 Ch**) [3] (80 Ash)
3. Why it is important to complete project successfully? [2] (**81 Ch**)

## 11.2 Communications planning process, communication requirement analysis, organizing and conducting effective meeting

1. Explain briefly on project communication planning. [3] (81 Ash)
2. Explain the hazards of communication error in a project. [2] (82 Ka, 79 Ash)
3. Find the no. of communication channels among 7 people. [2] (**81 Ch**)
4. Compare Formal versus informal communication in PM. [3] (**80 Ch**)
5. As a project manager how do you conduct meeting effectively? [2] (80 Ash) [3] (**79 Ch**)
   |→What are the steps to conduct effective meeting? [2] (81 Ash)
6. As project manager, how do you write effective email? [2+2] (**78 Ch**)

## 11.3 Information distribution process

## 11.4 Performance reporting process, integrated reporting system

1. Why reporting system is required in project? [2] (79 Ash) [3] (82 Ka)

# 12 Project Risk Management

(4 Hours / 7 Marks)

## 12.1 Understanding Risk, Project risk

1. What are the different sources of risk in IT project? [2+3] (79 Ch)
2. What is project risk? [1] (**78 Ch**)

## 12.2 Risk Management planning process, risk management plan

1. What is project risk management? [2] (**81 Ch, 79 Ch**)
2. Explain about risk management plan. [4] (**78 Ch**)
3. Why risk management is an essential part of project management? [2] (80 Ash)

## 12.3 Risk identification, risk identification techniques

1. What are the risk identification techniques? [3] (**81 Ch**)
2. Discuss briefly on risk identification techniques. [2] (**80 Ch**)
   |→in ICT project. [3] (80 Ash)
3. Create a sample risk register of any ICT related project. [3] (**80 Ch**)

## 12.4 Qualitative risk analysis process

1. Short notes on: Quantitative Risk Analysis [3] (**81 Ch**)
2. Describe different methods of qualitative analysis of risk. [4] (79 Ash)
3. (Assumed) Short notes on SWOT Analysis. [3] (82 Ka, 80 Ash)

## 12.5 Quantitative risk analysis process, modeling techniques

1. Describe different methods of quantitative analysis of risk. [4] (79 Ash)

## 12.6 Risk response planning, resolution of risk

## 12.7 Strategies for negative risks or threats, strategies for positive risks or opportunities

## 12.8 Risk monitoring and control process

## 12.9 Sensitivity

1. Short notes on: Tornado Analysis. [3] (81 Ash)

# 13 Project Procurement Management

(3 Hours / 5 Marks)

## 13.1 Procurement management process flow

1. What are the benefits of procurement management? [2] (82 Ash)
2. What do you mean by project procurement management? [2] (**81 Ch, 80 Ch**, 82 Ka, 80 Ash)
3. Explain about procurement management process flow. [2+3] (**79 Ch**)

## 13.2 Plan purchases and acquisition process, enterprise environmental factor

## 13.3 Organizational process assets

1. Short notes on: Organizational Process Asset. [3] (82 Ash)

## 13.4 Plan contracting process, standard forms, evaluation criteria

## 13.5 Processes

1. Explain the different processes adopted in the procurement management?
   [3] (**80 Ch**) [4] (**81 Ch**, 82 Ka, 80 Ash)
2. Discuss the process of procurement in any project. [5] (79 Ash)

### 13.5.1 Request seller response process

### 13.5.2 Select seller process

### 13.5.3 Contract Administration process

1. (Assumed) List out the components of bidding document. [3] (81 Ash)

### 13.5.4 Contract closure process

# 14 Developing Custom Processes for IT Projects

(3 Hours / 5 Marks)

## 14.1 Developing IT project management methodology

1. (Assumed) Discuss briefly the challenges in IT project in Nepal. [4] (**80 Ch**)
2. (Assumed) What are the reasons for failure of the ICT projects? [5] (**78 Ch**)
3. (Assumed) What is system development methodology? [2] (81 Ash)
4. (Assumed) Mention atleast three different methods and compare with Agile development methodology. [4] (81 Ash)

## 14.2 Moving forward with customized management process

1. Short notes on: Need custom processes for IT project [4] (**79 Ch**)[2] (**78 Ch**)
   |→Why do you need custom processes for IT Projects? [4] (80 Ash)
2. Small IT companies can't follow all the suggestions as per PMI, so they can develop their own custom processes for managing project. Do you agree? Discuss. [3] (79 Ash)

## 14.3 Certified associate in project management

## 14.4 Project management maturity

1. (Assumed) Write short notes on: PMMM. [3] (82 Ka)

## 14.5 Promoting project Excellency through awards and assessment

## 14.6 Certification process flow

## 14.7 Code of ethics

1. Write short notes on ICT Code of Ethics. [3] (82 Ka)

## 14.8 Future trends

1. Short notes on: Future trend of IT projects [4] (**79 Ch**)
2. Short notes on: Possible impact of AGI on Software project management. [3] (81 Ash)

# 15 Balanced Scorecard and ICT Project Management

(1 Hours / 2 Marks)

1. Write short notes on: Balanced Scorecard. [2] (**78 Ch**) [3] (**81 Ch**, 82 Ka, 81 Ash, 80 Ash)
2. Compare: KPI versus Balanced scorecard. [4] (**80 Ch**)